

CC0007 SCIENCE & TECHNOLOGY FOR HUMANITY

Course Aims

This course aims to inspire a long-lasting mindset of awareness, critical thinking, curiosity, and collaboration across disciplines through the lens of contemporary and near-future challenges for human communities in relation to scientific and technological innovations. You will learn to perceive and analyse the potential benefits and costs of scientific/technological innovations and applications from different perspectives and on different scales. You can then use these skills to identify real-life challenges and to propose solutions.

Intended Learning Outcomes (ILO)

Upon the successful completion of this course, you (as a student) will be able to:

1. Describe the basic scientific/technical principles of the featured cutting-edge research/innovations; the kinds of human problems/challenges that these are meant to address; and the additional problems/challenges posed in and by their application.
2. Describe and analyse connections between three general types of perspective on the challenges of innovation and application: scientific/technical, business, and humanistic/social scientific perspectives.
3. Identify, critique, and evaluate ethical, financial, organizational, legal, social, and cultural components surrounding the application of technological innovations.
4. Identify and present possible innovations and solutions for their application that would achieve a consensual balance between scientific-business-social concerns.
5. Collaborate effectively on project development in cross-disciplinary teams.

Course Requirements

There are no final examinations in this course. However, this course involves reading, writing, discussion, and groupwork. You will need to watch the pre-recorded lectures before you come to class, and be prepared to participate in the quizzes, activities, and discussion during class, as they will account for a significant portion of your grade. Several course components will be assessed on a group level, so you will have to work with your group members throughout the semester.

Consistent effort throughout the semester is required, and students are highly encouraged to proactively consult their instructors if they want more feedback on their progress in the course or encounter any other learning issues. The following are the assessment components for this course:

Component	Weighting	Type	Notes	Assessment rubrics
1. In-class quizzes	15% iRA 5% tRA 2% others	Individual	4 out of 5 iRA/TRA, drop 1 lowest score	See Appendix 1
2. Class participation	10%	Team/Individual	5% team + 5% individual	See Appendix 1
3. Proposal	20%	Team		See Appendix 2
4. Poster and presentation	25%	Team	10% poster + 15% oral presentation	See Appendix 3, 4
5. Reflection paper	20%	Individual		See Appendix 5
6. Reflective exercise	3%	Individual	Pre and post course survey conducted by ICC Office	See Appendix 6

Readings and References

Course materials will be provided during the course on the online learning platform.

Course Policies and Student Responsibilities

Students are expected to:

1. Complete all course activities.
2. Submit all assignments on time.

Course Policy on Absence

You are required to attend each tutorial session. Due to the nature of group activities, it is difficult to provide any online version of make-up tutorials.

If you are unable to attend class due to a valid reason, go to your **Class site** in NTULearn and fill out the form for MC Submission. Documentary proof of reason must be provided – upload your proof on the submission site. For absences other than medical reasons, we will evaluate your reason on a case-by-case basis. You are also encouraged to contact your group members for their understanding in advance if possible.

For missing tutorial sessions, you are encouraged to catch up with your group members for any missed group assignment and to make it up to your group by contributing more to their next assignment. An alternative suggestion is for the group to invite the affected students to join group activity via Zoom if the symptom of the affected students is mild. It is up to individual groups.

As the iRAs and tRAs are administered in a **closed-book format**, they must be attempted **in person**. Students who are late or absent for their tutorials without any valid excuse are not allowed to take any make-up quizzes, and their score for the missed quiz will be marked zero. **Any attempt to do the quizzes while not in class or any attempt to help others do so is tantamount to cheating and will be dealt with accordingly.**

Academic Integrity

Good academic work depends on honesty and ethical behaviour. The quality of your work as a student relies on adhering to the principles of academic integrity and to the NTU Honour Code, a set of values shared by the whole university community. Truth, Trust and Justice are at the core of NTU's shared values.

As a student, it is important that you recognize your responsibilities in understanding and applying the principles of academic integrity in all the work you do at NTU. **Not knowing what is involved in maintaining academic integrity does not excuse academic dishonesty.** You need to actively equip yourself with strategies to avoid all forms of academic dishonesty, including plagiarism, academic fraud, collusion, and cheating. If you are uncertain of the definitions of any of these terms, you should go to the [academic integrity website](#) for more information. Consult your instructors if you need any clarification about the requirements of academic integrity in the course.

Course Policy on the Use of Generative AI

NTU's position is that we should equip students with the knowledge and skills to use these AI technologies productively in an ethical and critical manner, and to help them sharpen their cognitive skills by synthesizing ideas, performing in-depth analysis, and working creatively. Students are expected to continue to practice high standards of academic and professional honesty and integrity.

In CC0007 you may use GAI programs (e.g., ChatGPT) to help in some aspects of your work. However, you should be aware that the materials generated by GAI programs may contain inaccuracies and can

be incomplete. Please be aware that the use of GAI may also stifle the development of your skills in writing as well creative and critical thinking.

Contributions from GAI programs must be properly quoted and cited every time they are used. **Failure to do so is considered a violation of academic integrity.** Please look [here](#) to see how to cite GAI content.

Please make proper declaration in the “Declaration of Academic Integrity” if you employ any GAI programs to help in your work and submit the supporting document(s) as required.

For CC0007:

- In-class activities
 - GAI is permitted for idea generation.
 - However, you should not read off GAI output directly during sharing/discussion.
 - Participation scoring will be based on quality of teamwork, team-discussion, class presentation, Q&A, etc.
- Poster and Proposal group project work
 - GAI is permitted for idea generation.
 - However, your original work with no AI content must be provided in an appendix.
 - Failure to do so will be in breach of your signed declaration and detection of GAI-usage will be penalized.
- Individual reflection papers
 - Since this should be a reflection of your own course experience, we see no reason for you to use GAI.
 - No AI content is allowed in the final submission. Detected usage will be reviewed and penalized.

Course Instructors

This is a large interdisciplinary course, which involves many instructors from different schools in NTU. If you have any queries, please contact the in-class instructor assigned to your group. Refer to the instructor contact list for contact information of individual instructors.

For problems with the learning platform (e.g., videos not working, dead links), please contact: [CITS](#)

If you are experiencing illness, personal distress, or other mental health-related issues which may affect your performance in the course, please reach out to your cluster instructor and/or lead instructor. You can also find professional counselling services provided by the [University Wellbeing Office](#).

Weekly Schedule

Week #	Week of	Topic	Task/Notes
1	13 Jan	Course Introduction	
2	20 Jan	Ageing I (STEM)	iRA/tRA (not graded)
3	27 Jan	E-learning week	
4	3 Feb	Ageing II (Business/Governance)	iRA/tRA
5	10 Feb	Ageing III (Social Science/Humanities)	iRA/tRA
6	17 Feb	Ageing IV (Groupwork consultation)	
7	24 Feb	Ageing V (Panel with Industry Experts)	Proposal due BEFORE tutorial
Recess Week			Mock Teamwork Peer Evaluation opens
8	10 Mar	Artificial Intelligence I (STEM)	iRA/tRA
9	17 Mar	Artificial Intelligence II (Business/Governance)	iRA/tRA
10	24 Mar	Artificial Intelligence III (Social Science/Humanities)	iRA/tRA
11	31 Mar	Artificial Intelligence IV (Panel with Industry Experts)	
12	7 Apr	Introduction to Blockchain (Panel with Industry Experts)	Teamwork Peer Evaluation opens
13	14 Apr	Poster Presentation	Poster due BEFORE tutorial Reflection Paper due 23:59 on your presentation day Teamwork Peer Evaluation due Sunday 23:59
14	21 Apr	Poster Presentation II	For Monday and Friday classes only

Appendix 1:

Assessment Criteria for Pre-lesson Learning (iRAs, tRAs, other activities, and class participation)

Pre-lesson learning is a prerequisite for participation in team-based learning activities during tutorials. It will be measured through individual and team quizzes, and other activities, and accounts for 22% of your grade. This consists of your iRA grade (15%), tRA grade (5%), and other in-class exercises on select weeks (2%).

To do well on this assessment you need to prepare for tutorials by watching the lectures beforehand. Your readiness will be assessed during most classes via a summative test in each session in the form of Individual Readiness Assessment (iRA) and Team Readiness Assessment (tRA). These will normally consist of 5 multiple-choice questions each. The questions will cover the material from the week's content and are designed to help you practice applying your knowledge and provide feedback on any gaps in your knowledge.

Note: The iRAs and tRAs are administered in a **closed-book format** and must be attempted **in person**. Students who are late or absent without any valid excuse are not allowed to take make-ups. **Any attempt to do the quizzes while not in class or any attempt to help others do so is tantamount to cheating and will be dealt with accordingly. Any attempt to refer to course materials or external references during and in-between quizzes will be dealt with similarly.**

Preparing for each class will also help you in another component of the course: in-class participation (10%). There will be group activities during the tutorial, and you are expected to participate actively in them. You also need to communicate actively with your team members in the spirit of cooperation. Speaking, listening, and respecting other viewpoints are all parts of this. This will be assessed on both a group level as well as an individual level to account for within-group differences.

	<i>Exemplary</i>	<i>Proficient</i>	<i>Developing</i>	<i>Unsatisfactory</i>
Assessment criteria for in-class participation (individual) – 5%				
Participation effort	Shows interest and effort to participate.	Moderate interest and effort to participate.	Occasional interest and effort to participate.	Little to no interest and effort to participate.
Quality of Participation	Input reflects excellent achievement of the learning outcomes.	Input is adequate and reflects adequate achievement of the learning outcomes.	Input is limited and reflects some achievement of the learning outcomes.	Input is minimal and reflects little to no achievement of the learning outcomes.
Assessment criteria for in-class participation (group) – 5%				
Frequency of group participation	Frequent contributions in class.	Moderate contributions in class.	Occasional contributions in class.	Little to no contributions.
Quality of groupwork	Excellent effort to meet requirements of the task.	Generally meets requirements of the task.	Some attempt to meet requirements of the task.	Minimal effort to meet requirements of the task.

Appendix 2: Assessment Criteria for Group Proposal

For this assessment you as a group are to hand in a written proposal (800-word limit; figures and other creative presentations are all welcome). This accounts for 20% of your grade. You will work in teams of 5 to 6. To do well on this team assessment, it is necessary for you as a team to demonstrate positive interdependence and teamwork. In principle, you will receive the same marks as your team. However, your individual score may vary depending on peer evaluation based on the Teamwork Evaluation Criteria (Appendix 4). The 800-word limit excludes titles for figures and tables, paragraph headers, in-text citations, and reference list (if any).

The group proposal assignment is due **BEFORE your tutorial in Week 7** in NTULearn. Please submit the “Declaration of Academic Integrity” to the appropriate portal.

Your proposal should include the following information:

- Identify a key problem or challenge that will be addressed.
- Identify your key technology/innovation for addressing the problem.
- Explain how your project can address the problem or challenge that you have identified.

	<i>Exemplary</i>	<i>Proficient</i>	<i>Developing</i>
Problem Identification	• The problem identified is significant and/or prevalent yet has been ignored or not been adequately addressed. • Provides compelling evidence and justification from multiple sources to support its significance. • Clearly defines the scope and boundaries of the problem. • The problem identified is relevant to the theme and markedly different from what has been discussed in class (i.e., not just a rehash of examples in class).	• The problem identified is somewhat significant and/or prevalent, and there have been attempts to address this problem. • Provides adequate evidence and justification from credible sources to support its significance. • Defines the scope and boundaries of the problem with minor ambiguities. • The problem identified is somewhat relevant to the theme and is somewhat similar to examples discussed in class.	• The problem identified is not very significant and/or prevalent and has already been adequately addressed. • Evidence and justification for significance is weak or lacking. • Scope and boundaries of the problem are unclear or poorly defined. • The problem identified is not very related to the theme or is the same as the examples discussed in class.
Critical Thinking	• The proposal displays perceptive insight and innovativeness. • There is a logical analysis that demonstrates clear understanding of relevant concepts. • There is an excellent demonstration of understanding and application of course material to analyse key points. • The analysis is relevant and shows depth. • The proposal is convincing and well-justified.	• The proposal displays some insight or some innovative components. • The analysis is generally sound, but there are lapses in logic or understanding. • There is some attempt at demonstration of understanding and application of course material to analyse key points. • The analysis is relevant but lacks depth.	• The proposal displays little innovativeness and/or lacks insight. • Analysis is illogical and reflects inadequate understanding of relevant concepts. • No attempt is made at demonstration of understanding and application of course material to analyse key points. • The analysis lacks relevance and depth.

	• Raises complex questions that provoke deeper understanding of the issue.	• The proposal is somewhat convincing and well-justified. • Raises basic questions that lead to further exploration of the problem.	• The proposal is not convincing, or not well-justified. • Raises superficial questions that do not contribute to deeper understanding.
Problem Solving	• The proposal demonstrates thoughtful consideration of the identified problem and the proposed solution from technical, business/governance, and social angles. • There is a convincing link between the identified problem and the proposed solution. • The solution demonstrates significant value over existing solutions. • Accurate details are used to support the proposed solution. • Anticipates potential challenges/limitations and proposes mitigations.	• The proposal demonstrates some consideration of the identified problem and the proposed solution from technical, business/governance, and social angles. • The link between the identified problem and the proposed solution is somewhat established. • The solution has some added value over existing solutions. • The analysis is relevant but lacks depth. • Details are included, but accuracy and relevance are somewhat questionable. • Identifies some potential challenges/limitations but lacks clear mitigation strategies.	• The proposal lacks consideration of the identified problem and the proposed solution from technical, business/governance, and social angles. • The link between the identified problem and the proposed solution is ambiguous or not adequately established. • There is little or no value-add over existing solutions. • Details used to support proposed solution are missing or irrelevant. • Superficial or little consideration of potential challenges/limitations.
Organisation, Presentation and Style	• Content is well-structured, logical, and flow is natural. • Paragraphs are well focused and coherent. • There are clear transitions between sections. • Content is clear and easy to understand. • Proposal is free of unnecessary detail. • Good use of visuals to enhance content and/or comprehensibility. • All necessary content and visuals are cited. • Writing is professional and articulate, free from grammatical errors and typos. • Delivers a polished and engaging write up that effectively communicates the project's key points and engages the audience	• Content is somewhat structured and there is some awkwardness in flow. • Paragraphs occasionally lack focus or coherence. • There are minor inconsistencies in transitions between sections. • Content is mostly clear and somewhat easy to understand. • Proposal is somewhat free of unnecessary detail. • Visuals are somewhat helpful in enhancing content and/or comprehensibility. • Most content and visuals have citations. • Writing is mostly clear and grammatically correct with some minor errors. • Delivers a write up with some clarity and engagement but may have minor flaws or inconsistencies.	• Content is poorly structured, disjointed, and/or flows poorly. • Paragraphs lack coherence or focus. • There is a lack of clear transitions between sections. • Content is vague or difficult to understand. • Proposal has details unrelated to the topic. • Visuals are poorly selected and not very helpful in enhancing content and/or comprehensibility. • Most content and visuals lack citations. • Writing is unclear, confusing, with many grammatical and spelling errors. • Delivers a confusing or poorly engaging write up that lacks clarity and focus.

Appendix 3: Assessment Criteria for Poster and Presentation

This assignment accounts for 25% of your grade (10% for poster, 15% for oral presentation).

The final submission will be a single-page digital poster which will be presented for a maximum length of ten minutes. You DO NOT need to print the poster. Prepare it as a PDF for presenting in class. Your group can decide on the number of presenters in the group. Not everyone needs to present.

To do well on this team-based assessment, it is necessary for you as a group to demonstrate positive interdependence and group work. In principle, you will receive the same marks as your group. However, your individual score may vary based on the Teamwork Evaluation Criteria (Appendix 4).

A soft copy of your poster should be submitted to NTULearn **BEFORE your presentation session**. Presentation will be done in class during your tutorial. Please submit the “Declaration of Academic Integrity” to the appropriate portal.

Your poster should include the following information:

- Identify a key problem or challenge that will be addressed. You can focus on any sector of your choice, **excluding** the following:
 - Cybersecurity
 - Fake news
 - Intellectual Property related issues
 - Data ethics related issues
- Explain the significance of the problem or challenge.
- Propose a key technology/innovation for addressing the problem. This should involve AI and machine learning.
- Identify the intended target audience of your proposed technology/innovation.
- State the feasibility and effectiveness of your project in addressing the identified problem or challenge.

Also consider how to make your poster attractive in terms of design, layout, and neatness. As this is meant to be a static poster, no videos or other animations should be included. The poster should be self-explanatory (i.e., the reader does not need additional input to understand it)

	<i>Exemplary</i>	<i>Proficient</i>	<i>Developing</i>
Problem Identification	• The problem identified is significant and/or prevalent yet has been ignored or not been adequately addressed. • Provides compelling evidence and justification from multiple sources to support its significance. • Clearly defines the scope and boundaries of the problem. • The problem identified is relevant to the theme and markedly different from what has been discussed in class (i.e., not just a rehash of examples in class).	• The problem identified is somewhat significant and/or prevalent, and there have been attempts to address this problem. • Provides adequate evidence and justification from credible sources to support its significance. • Defines the scope and boundaries of the problem with minor ambiguities. • The problem identified is somewhat relevant to the theme and is somewhat similar to examples discussed in class.	• The problem identified is not very significant and/or prevalent and has already been adequately addressed. • Evidence and justification for significance is weak or lacking. • Scope and boundaries of the problem are unclear or poorly defined. • The problem identified is not very related to the theme or is the same as the examples discussed in class.
Critical Thinking	• The proposed idea displays perceptive insight and innovativeness.	• The proposed idea displays some insight or some innovative components.	• The proposed idea displays little innovativeness and/or lacks insight.

	• There is a logical analysis that demonstrates clear understanding of relevant concepts. • There is an excellent demonstration of understanding and application of course material to analyse key points. • The analysis is relevant and shows depth. • The proposed idea is convincing and well-justified. • Raises complex questions that provoke deeper understanding of the issue.	• The analysis is generally sound, but there are lapses in logic or understanding. • There is some attempt at demonstration of understanding and application of course material to analyse key points. • The analysis is relevant but lacks depth. • The proposed idea is somewhat convincing and well-justified. • Raises basic questions that lead to further exploration of the problem.	• Analysis is illogical and reflects inadequate understanding of relevant concepts. • No attempt is made at demonstration of understanding and application of course material to analyse key points. • The analysis lacks relevance and depth. • The proposed idea is not convincing, or not well-justified. • Raises superficial questions that do not contribute to deeper understanding.
Problem Solving	• The proposal demonstrates thoughtful consideration of the identified problem and the proposed solution from technical, business/governance, and social angles. • There is a convincing link between the identified problem and the proposed solution. • The solution demonstrates significant value over existing solutions. • Accurate details are used to support the proposed solution. • Anticipates potential challenges/limitations and proposes mitigations.	• The proposal demonstrates some consideration of the identified problem and the proposed solution from technical, business/governance, and social angles. • The link between the identified problem and the proposed solution is somewhat established. • The solution has some added value over existing solutions. • The analysis is relevant but lacks depth. • Details are included, but accuracy and relevance are somewhat questionable. • Identifies some potential challenges/limitations but lacks clear mitigation strategies.	• The proposal lacks consideration of the identified problem and the proposed solution from technical, business/governance, and social angles. • The link between the identified problem and the proposed solution is ambiguous or not adequately established. • There is little or no value-add over existing solutions. • Details used to support proposed solution are missing or irrelevant. • Superficial or little consideration of potential challenges/limitations.
Organisation, Presentation and Style	• Content is well-structured, logical, and flow is natural. • Poster is self-explanatory and sufficient, without excessive unnecessary detail. • Good use of visuals to enhance content and/or comprehensibility. • All necessary content and visuals are cited. • All key information on the poster is visible from 1m away.	• Content is somewhat structured and there is some awkwardness in flow. • Poster is somewhat self-explanatory and/or has some redundancy. • Visuals are somewhat helpful in enhancing content and/or comprehensibility. • Most content and visuals have citations. • Almost all key information on the poster is visible from 1m away.	• Content is poorly structured, disjointed, and/or flows poorly. • Poster is not self-explanatory and/or has much redundancy. • Visuals are poorly selected and not very helpful in enhancing content and/or comprehensibility. • Most content and visuals lack citations. • Most of the information on the poster is unclear or too small.
Oral Presentation	• Presentation was engaging and pitched appropriately for intended target audience. • An ideal volume and speed were used for the presentation.	• Presentation was somewhat engaging and was somewhat pitched appropriately for intended target audience. • Volume and speed used for the presentation were somewhat appropriate.	• Presentation was not very engaging and was not pitched very appropriately for intended target audience. • Volume and speed used for the presentation were not very appropriate.

Appendix 4: Peer Evaluation

There will be two rounds of peer evaluation:

1. *Formative/qualitative feedback (ungraded) in week 7.* All teams should start by discussing and agreeing on their responsibilities including expected team behaviours and deliverables of each member. Every team member will have the opportunity to give qualitative feedback to each other in Week 7 (after the completion of the first proposal). Although this round of feedback will not affect students' final average peer evaluation score, it serves as an avenue to acknowledge areas where members have done well. It is also a chance for groupmates to suggest possible areas for improvement to members whose engagement or contributions fall short of the group's expectations. The goal is to encourage all members to improve the working dynamics for the presentation in the second half of the semester. It will not be fair to point out any member's uncollaborative behaviours only at the end of the course if a member has not been fulfilling his/her role and responsibility from the beginning. Students should avoid identifying themselves in any manner when providing qualitative comments to their team members. If there are any concerns regarding group dynamics, please contact your current cluster instructor immediately upon receiving the qualitative comments.
2. *Peer evaluation in Week 13.* The final peer evaluation will take place at the end of the semester (Week 13), during which you will be rated holistically for your contribution to all groupwork during the semester. Please note that during the peer evaluation process, students will be required to provide a written justification for their quantitative ratings of their groupmates. All team members are expected to complete the peer evaluation for all other members in the team (i.e., self-assessment is not required); see evaluation criteria and rating descriptors in the following page. The mark deduction, if any, will be imposed on assessment components 2, 3, and 4 as listed on Page 1 of the syllabus.
3. Please complete the peer evaluation through NTULearn (under your respective clusters) for **ALL** your team members for the team assignment by the given deadline. Should a student fail to complete the teamwork evaluation of all other members in his/her group, 5% will be deducted from the student's final mark for assessment components 2, 3, and 4.

NOTES:

1. All assessments and qualitative comments are confidential, and we expect you to be constructive and fair in all your feedback and ratings. Team members will be able to see the compiled qualitative comments given by their team members given at mid semester but the average peer evaluation score only at end semester.
2. You will be able to access your average peer evaluation score from the start of Week 14. If you think that your score does not commensurate with your contribution to the group projects, you may submit an appeal to the *appeal portal on NTULearn before the given deadline* to share your concerns together with all supporting evidence (e.g., draft work, group email correspondence, chat messages, etc.). Otherwise, your score will be considered final. Based on the veracity of a student's explanation and submitted evidence (including other collected information), the course coordinators and instructor may maintain or adjust a student's average peer evaluation score accordingly. Please note that students' emails that have no supporting evidence will be dismissed without further consideration.
3. Please note that the teaching team reserves the right to adjust students' final team assignment marks based on additional considerations including gathered information, certified special education needs (SEN), and medical diagnosis.
4. If you have certified SEN or medical concerns that might affect your contributions and collaboration, but you are having trouble communicating your situation to your groupmates, please approach your tutor for a chat in advance.

Peer Evaluation Criteria

Criteria (Weights)	Score from 1 to 9 ^{1,2} (1: Never; 3: Rarely; 5: Occasionally; 7: Frequently; 9: Always)				
(For 6-members team)	Member A	Member B	Member C	Member D	Member E
Member name					
a. Fulfilling one's responsibilities duly (15%)					
Behaved responsibly—such as attend meetings punctually and regularly; participate in discussion; complete assigned tasks/roles punctually.	Score from 1 to 9				
	Qualitative comments/reasons				
b. Fulfilling one's responsibilities effectively (25%)					
Behaved and contributed effectively—such as quality of work produced; creativity of ideas; extensiveness of research and thinking.	Score from 1 to 9				
	Qualitative comments/reasons				
c. Managing interpersonal relationships (30%)					
Listened attentively to and sought inputs from others; helped team resolve conflicts and achieved common understanding to function effectively; promoted respect for others and differences; fostered camaraderie.	Score from 1 to 9				
	Qualitative comments/reasons				
d. Providing support to others to achieve goals (30%)					
Behaved fairly and ethically—such as sharing responsibilities and giving credits. Exhibited group citizenship behavior—such as helping others to learn and complete their work through guidance and encouragement; standing up for others when needed.	Score from 1 to 9				
	Qualitative comments/reasons				

¹ Score of 1 should be given only when a team member does not really deserve to be awarded any mark for the team assignment (i.e., zero mark) because the member either has not or has barely participated and/or contributed to the team assignment in any meaningful manner.

² See detailed score descriptions for each criterion in Appendix 3b.

Peer Evaluation Criteria – Score Descriptions

Please use the descriptors in to guide your evaluations. For example, if the peer exceeds the descriptor for “1” but does not yet meet the threshold described in “3”, then select “2”

a. Fulfilling one's responsibilities duly (15%)									
	1	2	3	4	5	6	7	8	9
	Never		Rarely		Occasionally		Frequently		Always
Behaved responsibly--such as attend meetings punctually and regularly; participate in discussion; complete assigned tasks/roles punctually.	Did not participate actively in team meetings at all (even though he/she might still be present.) Did not contribute any useful ideas that were relevant for getting the project completed. Did not submit any or usable work on time and never did so without prompting.		Rarely participated actively in team meetings. Rarely contributed useful ideas that were relevant for getting the project completed. Rarely submitted work on time and did so rarely without prompting.		Occasionally participated actively in team meetings. Occasionally contributed useful ideas that were relevant for getting the project completed. Occasionally submitted work on time and did so occasionally without prompting.		Frequently participated actively in team meetings. Often contributed useful ideas that were relevant for getting the project completed. Frequently submitted assigned work on time without prompting.		Always participated actively in team meetings. Always contributed useful ideas that were relevant for getting the project completed. Always submitted assigned work on time without prompting.
b. Fulfilling one's responsibilities effectively (25%)									
	1	2	3	4	5	6	7	8	9
	Never		Rarely		Occasionally		Frequently		Always
Behaved and contributed effectively--such as quality of work produced; creativity of ideas; extensiveness of research and thinking.	Quality could not be evaluated because nothing relevant or usable was submitted.		Quality of work could be improved vastly; required major revisions before submission. Work reflected superficial effort in completing the project.		Quality of work could be improved; required some revisions before submission. Work reflected some effort in completing the project.		Quality of work was of reasonably acceptable standards; required minor revisions before submission. Work reflected reasonable effort in completing the project.		Quality of work was of highly acceptable standards; required no revision before submission. Work reflected extensive effort in completing the project.

<i>c. Managing interpersonal relationships (30%)</i>									
	1	2	3	4	5	6	7	8	9
	Never		Rarely		Occasionally		Frequently		Always
Listened attentively to and sought inputs from others; helped team resolve conflicts and achieved common understanding to function effectively; promoted respect for others and differences; fostered camaraderie.	Never listened to others. Never attempted to help team resolve conflicts. Never attempted to achieve common understanding to function effectively as a team. Never promoted respect for others, with no attempt to foster camaraderie.		Rarely listened to others. Rarely attempted to help team resolve conflicts. Rarely attempted to achieve common understanding to function effectively as a team. Rarely promoted respect for others, with rare attempts to foster camaraderie.		Occasionally listened to others, with occasional attempts to help team resolve conflicts. Occasionally attempted to achieve common understanding to function effectively as a team. Occasionally promoted respect for others, with occasional attempts to foster camaraderie.		Frequently listened to others, with frequent attempts to help team resolve conflicts. Frequently attempted to achieve common understanding to function effectively as a team. Frequently promoted respect for others, with notable attempts to foster camaraderie.		Always listened to others, always attempted to help team resolve conflicts. Always attempted to achieve common understanding to function effectively as a team. Always promoted respect for others and demonstrated consistent attempts to foster camaraderie.
<i>d. Providing support to others to achieve goals (30%)</i>									
	1	2	3	4	5	6	7	8	9
	Never		Rarely		Occasionally		Frequently		Always
Behaved fairly and ethically—such as sharing responsibilities and giving credits. Exhibited group citizenship behaviour—such as helping others to learn and complete their work through guidance and encouragement; standing up for others when needed.	Never took on a fair share of responsibilities; never gave appropriate credits to others. Never attempted to help others learn; never provided guidance and encouragement. Never stood up for others when needed.		Rarely took on a fair share of responsibilities; rarely gave appropriate credits to others. Rarely attempted to help others learn; rarely provided guidance and encouragement. Rarely stood up for others when needed.		Occasionally took on a fair share of responsibilities; occasionally gave appropriate credits to others. Occasionally attempted to help others learn; occasionally provided guidance and encouragement. Occasionally stood up for others when needed.		Frequently took on a fair share of responsibilities; frequently gave appropriate credits to others. Frequently attempted to help others learn; frequently provided guidance and encouragement. Frequently stood up for others when needed.		Always took on a fair share of responsibilities; always gave appropriate credits to others. Always attempted to help others learn; always provided guidance and encouragement. Always stood up for others when needed.

Peer Evaluation Scores and Corresponding Deductions (if any)

Each member's average peer evaluation score (out of 100% in *Peerceptiv* and rounded-up to integer) given by his/her team members will be used to determine their final team assignment marks as follows:

Peerceptiv's final score (out of 100%)	Marks Deduction from components 2, 3, 4 (out of 100%)	Final Adjusted Marks (out of 100%)
≥ 72	0%	Original team assignment mark (M%)
≥ 61 to < 72	-5%	M% – 5%
≥ 44 to < 61	-10%	M% – 10%
≥ 33 to < 44	-15%	M% – 15%
> 11 to < 33	-20%	M% – 20%
< 11	-M%	0%

Note:

If a student fails to complete the peer evaluation for all his/her other team members, an additional 5% deduction will be imposed on the student's final team assignment marks.

Example:

1. If a team receives 88% and 80% for Proposal 1 and Poster respectively and Member X who has *completed* the peer evaluation for all other members in his/her team receives an average Peerceptiv score of 50% (ie, 4.5 out of a 9-point scale), Member X's adjusted scores for the team assignments will be 78% and 70% respectively (i.e., a deduction of 10% per assignment).
2. If a team receives 88% and 80% for Proposal 1 and Poster respectively and Member Y who has *NOT* completed the peer evaluation for all other members in his/her team receives an average Peerceptiv score of 50% (ie, 4.5 out of a 9-point scale), Member Y's adjusted scores for the team assignments will be 73% and 65% respectively (i.e., a deduction of 10% plus 5% per assignment).

Appendix 5: Individual Reflection Paper

For this assessment you are to hand in an individual reflection paper (800-word limit; figures and other creative presentations are all welcome). This assignment accounts for 20% of your grade. The 800-word limit excludes titles for figures and tables; paragraph headers; in-text citations and reference list (if any).

This paper gives you an opportunity to reflect on your learning journey during the module and the insights you gained from CC0007. Write a paper that discusses the impact of this course's experience on your adoption of an interdisciplinary perspective. The assignment is due **at 23:59 on your poster presentation day** in NTULearn. (For example, if your poster presentation falls on Monday, it would be due 23:59 that day).

Additional points to note:

- This is not English composition. We will not penalise for poor grammar, nor award higher marks for fancy language.
- This is a personal reflection based on your course experience, not the experience of your friend who has taken the course earlier.
- Given the above, there is no reason to use Generative AI for your work, so GAI tools are **strictly not permitted**.

Please submit the "Declaration of Academic Integrity" to the appropriate portal. The grading rubric is as follows:

	<i>Exemplary</i>	<i>Proficient</i>	<i>Developing</i>
Personal response to the course experience	• Provides a clear description of their intellectual and emotional response to the experience. • There is clear and thoughtful reflection. • The reflection is relevant to the course and shows depth and personalisation.	• Some clear and focused description of their feelings and thoughts. • There is some reflection but may be ambiguous or unfocused. • The reflection is beyond general descriptions of what was done in the course.	• Unclear and vague response to the experience. • Reflection is vague without any real conclusion. • The personal reflection is irrelevant if not missing entirely.
Impact of course experience on interdisciplinary perspective	• The reflection provides a clear analysis of the impact of the course experience on own interdisciplinary perspective. • Insightful assessment of the disciplinary/interdisciplinary dimensions of learning/teamwork in the course. • Relevant details are used to support the reflection.	• The reflection is somewhat able to describe the impact of the course experience on own interdisciplinary perspective. • Insightful assessment of the disciplinary/interdisciplinary dimensions of learning/teamwork in the course. • Details are included, but not always relevant.	• The reflection is unable to describe impact of the course experience on own interdisciplinary perspective. • Insightful assessment of the disciplinary/interdisciplinary dimensions of learning/teamwork in the course. • Details are missing or irrelevant.
Critical Thinking	• Reflection displays insight and depth of thought. • Able to make connections between course content/experience and real-world applications.	• The reflection displays some depth of thought. • Some attempts to make connections between course content/experience and real-world applications.	• The paper displays little depth of thought or is not insightful. • Inadequate attempts to make connections between course content/experience and real-world applications.
Organisation, Presentation and Style	• Content is well-structured, logical, and the flow is natural. • Paragraphs are well- focused and coherent	• Content is somewhat structured but there is some awkwardness in flow. • Paragraphs occasionally lack focus or coherence	• Content is poorly structured, disjointed, and/or flows poorly. • Paragraphs lack coherence or focus

Appendix 6: Reflection exercises (3%)

This assessment comprises two reflection activities which are administered by the ICC Office, and are **qualitatively different** from the Individual Reflection Paper that you will be writing.

The first one will be at the beginning of the semester (*pre-course reflection*, to be completed between Weeks 2 and 4), and the second one at the end of the semester (*post-course reflection*, to be completed between Weeks 11 and 13). These

Completion of either reflection exercise will earn you 1% in your course grade, and completion of *both* reflection exercises will earn you 3% in total in your course grade. The reflection exercises are designed to engage you in thinking about, and reflecting on, your learning in the course as research on learning has suggested such activities to be beneficial for goal setting and academic performance (e.g., Travers et al., 2015¹).

Alternative assignments for reflection exercises. If you prefer to complete an alternative written assignment, please follow the instructions provided in the announcement on the reflection exercises that the coordinators will be uploading in via NTULearn. Similarly, completion of one alternative assignment will earn you 1% in your course grade, and completion of both alternative assignments will earn you 3% in total in your course grade.

¹ [Travers, C. J., Morisano, D., & Locke, E. A. \(2015\). Self - reflection, growth goals, and academic outcomes: A qualitative study. *British journal of educational psychology*, 85\(2\), 224-241.](#)